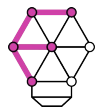


FAIR Principles for Research Software: From Principles to Practical Guidelines

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FAIR DATA
INNOVATIONS HUB



About Me

About Me

What we do at the FAIR Data Innovations Hub

We develop **software and standards** that help researchers easily prepare and share **FAIR, AI-ready data and other outputs**

All of our work is **free and open source**, supported by various organizations:



National Institutes
of Health



The
Navigation
Fund



Microsoft



About Me

What is FAIR?

FAIR Principles (2016): High-level recommendations to make research outputs Findable, Accessible, Interoperable, Reusable

To be Findable:

- F1. (meta)data are assigned a globally unique and persistent identifier
- F2. data are described with rich metadata (defined by R1 below)
- F3. metadata clearly and explicitly include the identifier of the data it describes
- F4. (meta)data are registered or indexed in a searchable resource

(and more)

About Me

FAIR data to FAIR software

Our main focus is on **developing software**
that makes it easy for researchers to **share FAIR data**

Random thought in 2021: We should make our software FAIR





Making Software FAIR

Making Software FAIR

FAIR4RS principles

FAIR Principles for Research Software or FAIR4RS Principles (2022)

FAIR Principles tailored for research software

F1. Software is assigned a globally unique and persistent identifier.
F1.1. Components of the software representing levels of granularity are assigned distinct identifiers.
F1.2. Different versions of the software are assigned distinct identifiers.
F2. Software is described with rich metadata.
F3. Metadata clearly and explicitly include the identifier of the software they describe.
F4. Metadata are FAIR, searchable and indexable.

(and more)

<https://doi.org/10.1038/s41597-022-01710-x>

Making Software FAIR

FAIR4RS Principles – why?



**Having a unique
identifier**



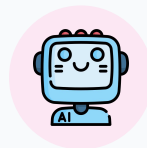
**Making your software
citable**



**Having extensive
metadata**



**Making your software
discoverable &
reusable**



**Focusing on machine
actionability**



**Making your software
ready for AI**

Making Software FAIR

The FAIR4RS principles are being broadly adopted



Policy Mandates

Many policies require compliance with the FAIR4RS principles
(e.g., Open Research Funders Group)



Financial Support

Grants for supporting compliance are available
(e.g., German Research Council - DFG)



Infrastructure & Tools

Infrastructure is being build for enabling compliance
(e.g., FAIR-IMPACT FAIR Metrics, CodeMeta)

Making Software FAIR

Challenge

The FAIR4RS Principles are **not actionable** by design

F1. Software is assigned a globally unique and persistent identifier.

How do I assign a unique identifier?

F1.1. Components of the software representing levels of granularity are assigned distinct identifiers.

F1.2. Different versions of the software are assigned distinct identifiers.

F2. Software is described with rich metadata.

What is rich metadata?

F3. Metadata clearly and explicitly include the identifier of the software they describe.

F4. Metadata are FAIR, searchable and indexable.

(and more)

Making Software FAIR

FAIR-BioRS guidelines

FAIR Biomedical Research Software (FAIR-BioRS) Guidelines (2023)

Minimal, actionable, step-by-step guidelines
for complying with each of the FAIR4RS principles

- ❑ Work from a version control system like GitHub
- ❑ Maintain a README
- ❑ Archive source code on a DOI-issuing repository like Zenodo
- ❑ Etc.

Making Software FAIR

Outstanding challenges despite the FAIR-BioRS guidelines

CHALLENGE 1

Time consuming and difficult to implement



TOOL

Codefair.io

CHALLENGE 2

Lack of inputs outside of biomedical research to be generalized



INITIATIVE

Actionable FAIR4RS Task Force



Actionable FAIR4RS Task Force

Actionable FAIR4RS Task Force

Background



ORIGIN

**Launched in Dec 2024 under the
Research Software Alliance (ReSA)**



CORE OBJECTIVE

**Establish actionable guidelines To
make any research software FAIR in
line with FAIR4RS principles**

Actionable FAIR4RS Task Force

Members



MEMBERS

~15

Active Members



GEOGRAPHY

Various Locations

USA, Canada, Germany, Spain,
UK, Netherland, etc.



EXPERTISE

Various Domains

Biomedical, Data Science,
Knowledge Representation, etc.

Email me to join! bpatel@calmi2.org

Actionable FAIR4RS Task Force

Methods

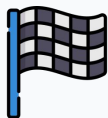


Everything is tracked through our repo's GitHub issues: <https://github.com/researchsoft/Actionable-FAIR4RS/issues>

Actionable FAIR4RS Task Force

Work in progress: draft guidelines

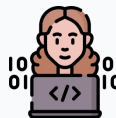
01



Before starting

- Select a license
- Work from a version controlled system platform (e.g. GitHub)

02



While developing the software

- Record dependencies (e.g. requirements.txt for Python code)
- Maintain metadata in codemeta.json, CITATION.cff

03



When releasing a new version

- Update metadata files with version specific PID
- Archive source code on a PID-issuing repository like Zenodo, Figshare, or Software Heritage

Actionable FAIR4RS Task Force

Timeline

October 2026

Release first draft
of the guidelines for
community
feedback

Q1 2027

Publish first
version of the
guidelines

Q2 2027

Maintain
guidelines



<https://www.researchsoft.org/tf-actionable-fair4rs>

Actionable FAIR4RS Task Force

Provide your comments now!

Help us by providing your comments on our draft guidelines!

bit.ly/FAIR4RS-guidelines

Scan QR Code to respond





Closing Comments

Closing Comments



Making research software FAIR
is essential



We are developing guidelines
to make it easier



You can all contribute by
providing feedback!
bit.ly/FAIR4RS-guidelines



Checkout [Codefair.io](https://codefair.io) and start
making your software FAIR today!

Thank You!

Contact & Connect



Email
bpatel@calmi2.org



Website
fairdataihub.org

Access Slides & Resources



Scan QR Code

Find all presentation
slides and related
resources

bit.ly/FAIR4RS-NASA